

SmolVLA: Dual-Stream Frozen CLIP Policy

Pretrained Vision-Language Backbone for Robotic Manipulation

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https://github.com/krazykarthik2/manual_vla

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1. Abstract

SmoIVLA is a compact Vision-Language-Action policy that leverages a frozen pretrained OpenAI CLIP ViT-B/32 backbone for multimodal perception, combined with lightweight cross-attention fusion and an Optimal Transport Conditional Flow Matching action decoder for trajectory generation on a 4-DOF Dobot Magician manipulator.

2. Architecture Overview

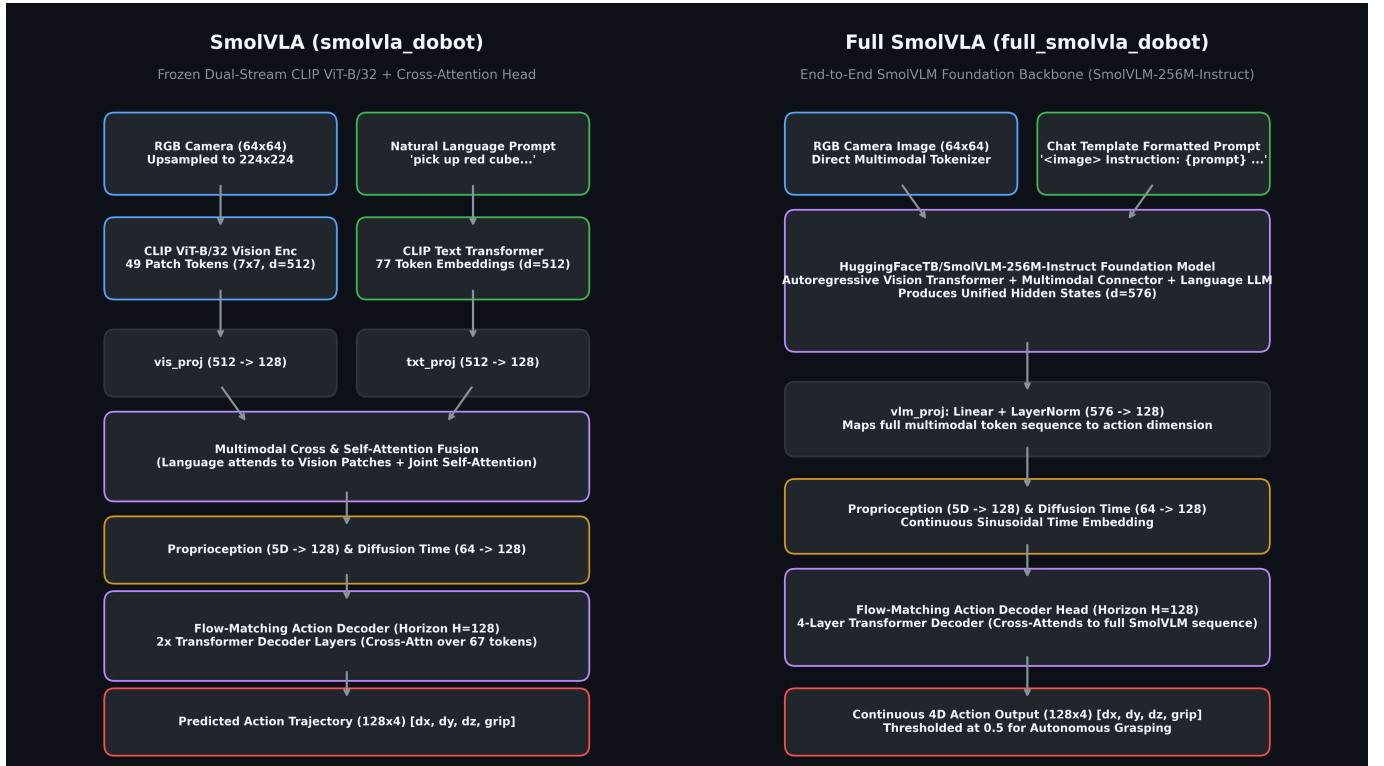


Figure 1: SmoIVLA dual-stream architecture with frozen CLIP ViT-B/32 backbone.

SmoIVLA follows a dual-stream design:

- Vision Stream: frozen CLIP ViT-B/32 visual encoder
- Language Stream: frozen CLIP text transformer
- Multimodal Fusion: trainable cross-attention + self-attention blocks
- Action Decoder: transformer-based flow matching decoder

3. Pretrained VLM Encoder (PretrainedVLMEncoder)

3.1 Vision Stream

Input: 64x64 RGB images upsampled to 224x224 with bilinear interpolation.

The image is normalized with CLIP's ImageNet statistics:

mean = [0.48145466, 0.4578275, 0.40821073]
std = [0.26862954, 0.26130258, 0.27577711]

Processing through CLIP's visual transformer:

1. Conv1 (patch projection): produces $7 \times 7 = 49$ spatial patch features
2. Prepend CLS token + add positional embeddings (50 tokens)
3. LayerNorm pre-processing
4. 12-layer ViT transformer
5. LayerNorm post-processing
6. Project through CLIP's visual projection matrix
7. L2-normalize

Output: 49 patch tokens in $\mathbb{R}^{49 \times 512}$ + 1 CLS embedding in $\mathbb{R}^{512}$.

All CLIP weights are frozen (`requires_grad=False`).

3.2 Language Stream

Input text instructions are tokenized into 77 BPE tokens.

Processing:

1. Token embedding lookup
2. Add positional embeddings
3. Cast to CLIP dtype (float16 on CUDA)
4. 12-layer text transformer
5. LayerNorm final
6. Project through `text_projection` matrix
7. L2-normalize

Output: 77 token embeddings in $\mathbb{R}^{77 \times 512}$. All weights frozen.

3.3 Spatial Grid Buffer

A 7×7 spatial coordinate grid in $[-1, 1]$ range is registered as a buffer, mapping each of the 49 visual patch tokens to a 2D spatial position for grounding and attention visualization.

4. Multimodal Fusion (MultimodalCrossAttentionBlock)

After projection from 512-dim to `d_model=128`, the fusion operates in two stages per block. There are 2 stacked fusion blocks.

4.1 Cross-Attention: Language attends to Vision

Query: `LayerNorm(text_tokens)` [B, 16, 128]

Key/Value: `LayerNorm(visual_patches)` [B, 49, 128]

Output: `cross_out` + residual `text_tokens`

This grounds language tokens in the visual scene, allowing the model to associate words like 'blue cube' with specific spatial patch regions.

4.2 Joint Self-Attention

The grounded text tokens and visual patches are concatenated:

`joint = [text_tokens || visual_patches] -> [B, 65, 128]`

Full self-attention is applied over the joint sequence, allowing bidirectional information flow between language and vision modalities.

4.3 Feed-Forward Network

A 2-layer MLP with GELU activation:

Linear(128 -> 256) -> GELU -> Linear(256 -> 128)

Applied with residual connection after LayerNorm.

5. SmoIVLA Backbone (SmoIVLABackbone)

The backbone class orchestrates the full forward pass:

1. Encode vision (frozen): 49 patches at 512-dim + CLS token
2. Encode text (frozen): 77 tokens at 512-dim, truncated to first 16
3. Compute zero-shot similarity heatmap:
 $\text{base_weights} = \text{einsum}(\text{'bld,bpd->blp'}, \text{text_feats}, \text{vis_patches})$
 $\text{base_probs} = \text{softmax}(\text{base_weights} * 5.0, \text{dim}=-1)$
4. Project both streams: 512 -> 128 via Linear + LayerNorm
5. Pass through 2 MultimodalCrossAttentionBlocks

Output: grounded text tokens [B, 16, 128], fused visual patches [B, 49, 128], and the pretrained attention heatmap [B, 77, 49].

6. Flow Matching Action Decoder

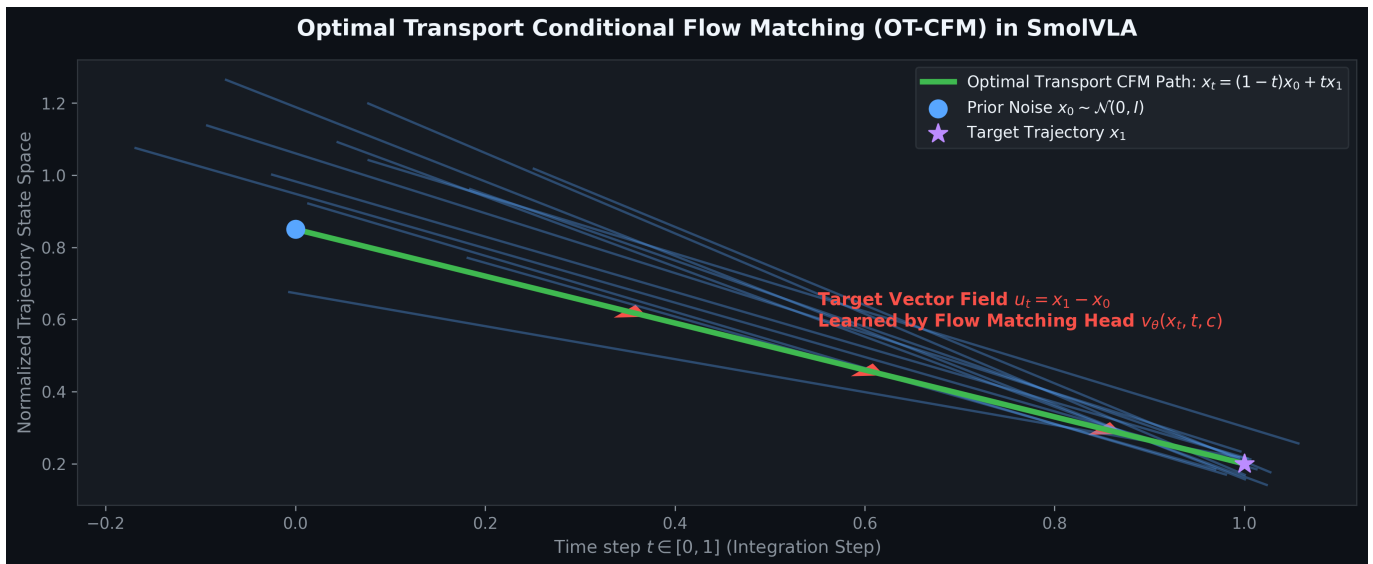


Figure 2: OT-CFM linear interpolation and velocity field.

The action decoder predicts 128-step continuous trajectories for 4 action dimensions (dx, dy, dz, grip) using Optimal Transport Conditional Flow Matching.

Formulation:

$x_t = (1-t)x_0 + t x_1$, where $x_0 \sim N(0, I)$ and x_1 is the demo trajectory

Target: $u_t = x_1 - x_0$ (constant velocity field)

Loss weights: $w = [1.2, 1.2, 1.5, 2.5]$ for $[x, y, z, \text{grip}]$

Inference: Euler integration from noise over 15 steps.

7. Visual Grounding Analysis

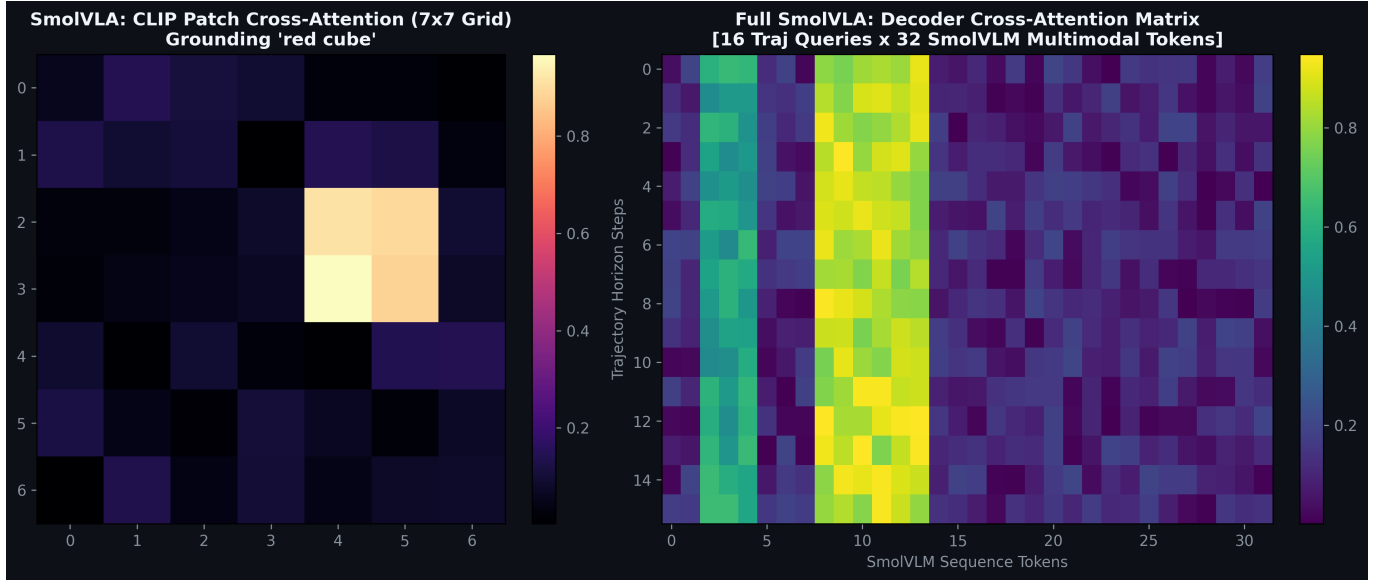


Figure 3: Cross-attention heatmaps and spatial grounding.